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FH6SmartShift (v1.1.0)

Virtual Dyno, Dragy Timer, Tuning Advisor & Shift Telemetry for Forza Horizon 6 & Forza Motorsport

Platform: SimHub | Games: FH6 & Forza Motorsport | .NET Framework 4.8 + WPF | License: GPLv3

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User & Technical Manual
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Build: v1.1.0 Dual-Game
    Sampling: 60 Hz UDP
```

1. OVERVIEW & CORE ARCHITECTURE (V1.1.0)

FH6SmartShift is a real-time C# telemetry plugin for SimHub and a racing dashboard suite engineered for the **Forza Horizon 6** and **Forza Motorsport** physics engines. Instead of relying on static redline percentages, the plugin constructs a live **401-band Virtual Dyno** (`50 RPM` resolution up to `20,000 RPM`) and calculates mathematical power curve intersections across Gears 1–10.

- **Dual-Game Engine (`FH6` & `FM`):** Automatically detects whether you are driving in *Forza Horizon 6* or *Forza Motorsport* (`[FH6SmartShift.ActiveGameMode]`). Prefixes *Forza Motorsport* profiles with `FM_` so databases never collide, and tightens tire slip thresholds (`0.42` in G1–3, `0.26` in G4+, `0.18 rad` slip angle) for track tires.
- **401-Band Self-Healing Virtual Dyno:** Samples engine power, torque, and turbo boost at wide-open throttle (`Accel >= 240` , ~94% throttle, `> 5000 W` , after 2 clean WOT frames) using an asymmetric filter (`alpha = 0.35` up / `0.08` down) and automatic 1–2 band pocket healing (`HealEngineMapPockets`).
- **Power Curve Intersection (`(P)`):** Compares current gear power (`pNow`) against next-gear landing power (`pNext` , including a `0.85` turbo boost-drop penalty if boost drops below 70%). Locks optimal shift when `pNext >= 1.015 * pNow` across 5 steps (`25 RPM` each) with `>= 78%` active landing zone density.
- **Fast & Bulletproof Ratio Learning (`(R)`):** Locks gear ratios in ~0.18 s after `12` tight-cluster samples (interquartile spread `< 1.2%`) from moderate throttle (`Accel >= 75`) when wheel slip is `< 0.08` . Enforces global ratio monotonicity across all gears (`1..10`) and auto-resets if tuning changes are detected.
- **Ghost Limiter Detection (`(L)`):** Detects hidden power cuts before nominal `EngineMaxRpm` , purges contaminated dyno bands, and unlocks automatically if the engine later revs past the ceiling.

2. NEW SUITE FEATURES IN V1.1.0

🏁 1. FH6 & Forza Motorsport + Active Gear Delegates

Auto-switches physics profiles between Horizon and Motorsport (`FM_` prefix) and exposes `CurrentGearColor` & `CurrentGearStatusText` for dynamic active-gear frames.

🏷️ 2. Custom Car & Build Naming

Assign custom names to specific `CarOrdinal` IDs or individual tunes via the `✏️ Name Car` button in the HTML report and save them to `FH6SmartShift_CarList.json` .

🔊 3. F1 / GT3 Audio Shift Beep

Zero-latency background tone (`2650 Hz` , `60 ms`) triggered the exact frame all 12 LEDs lock at `ShiftLightFlash` . Toggle via SimHub UI or `FH6SmartShift.ToggleShiftBeep` .

⏱️ 4. Built-In "Dragy" Timer

Sub-frame linearly interpolated acceleration timer measuring **0–100 km/h**, **100–200 km/h** (standing or rolling), and **0–200 km/h**, saving Personal Bests per car in JSON.

🔧 5. Gearbox Tuning Advisor

Embedded in `FH6SmartShift_GarageReport.html` . Analyzes RPM drops and landing RPM against peak torque and boost maps to recommend exact gear slider adjustments.

🖥️ 6. WPF Control Center & Profile Merge

Built-in SimHub settings tab with a **Reaction-Time Buffer Slider (0.10–0.40 s)**, plus single-car JSON export (`📤 Export Car`) and database merging.

3. QUICK INSTALLATION & FILE LOCATIONS

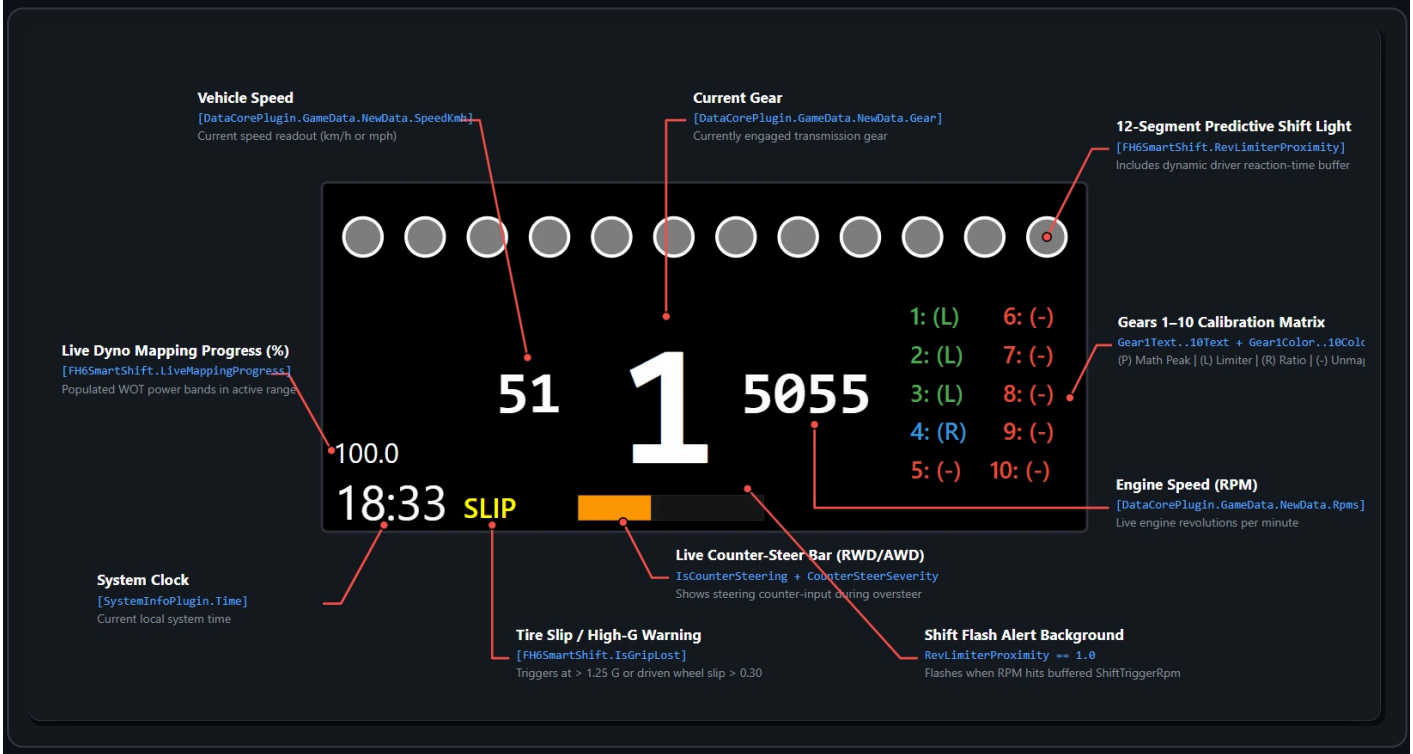
1. Close SimHub completely and copy `FH6SmartShift.dll` into your SimHub root folder (`C:\Program Files (x86)\SimHub\`).
2. Start SimHub, enable the plugin, and import `FH6SmartShift_Optimal.zip` / `DashTemplates.zip` into Dash Studio.
3. In **Forza Horizon 6** or **Forza Motorsport** (**HUD and Gameplay** → **Data Out**), enable UDP telemetry matching SimHub's IP and Port.
4. **Database, CarList & HTML Report:** Stored in `C:\Program Files (x86)\SimHub\PluginsData\Common\` (retains 3 daily backups `.bak_yyyyMMdd`).

4. DASHBOARD VISUAL REFERENCE (V1.1 OPTIMAL & CLASSIC LAYOUTS)

v1.1 Optimal MoTeC-Style Layout (1334×513 / 980×377): Features a sequential 60 Hz 12-LED bar (`LedActiveCount 1..12`), compact Dyno Status Pill (`DYNO OK 100%`), Smart Downshift Assistant (`▼ DOWN`), Live Shift Target Pill (`SHIFT @ 7520`), Multi-Warning Pill (`SLIP` / `OVRST!` / `BOTTOM!`), and the **3-Color Slide Level & Recovery Bar** (🔴 Red `▲ DEEPENING` | 🟠 Orange `● HOLDING` | 🟢 Green `◄ RECOVERING`).



Classic Telemetry Layout: Features the 12-segment predictive shift light bar, central Gear/Speed/RPM cluster, Gears 1–10 calibration matrix, and bottom slip/counter-steer indicators.



5. GEARS 1–10 CALIBRATION MATRIX STATES

State	Color / Hex	Actual Condition in C# Code (v1.1.0)
`N: (P)`	🟢 Green `#4CAF50`	**Peak Optimal (** `IsConfirmed && IsMathOptimal` **):** Confirmed power curve intersection with gear `N+1` (`pNext >= 1.015 * pNow` across 5 steps) and active landing zone density `>= 78%`.
`N: (L)`	🟢 Green `#4CAF50`	**Limiter Confirmed (** `IsConfirmed && !IsMathOptimal` **):** Confirmed optimal shift right at the true/ghost rev limiter (no power intersection before cutoff, or confirmed top gear).
`N: (L?)`	🟠 Orange `#F39C12`	**Edge / Unconfirmed Limit (** `TriggerIsEdge` **):** Provisional trigger (`actualLimiter - 250 RPM`) while the upper power band is still being populated.
`N: (P?)`	🟡 Yellow `#F1C40F`	**Unconfirmed Trigger:** Gear has a saved shift point (`ShiftTriggerRpm > 0`), awaiting full density confirmation.
`N: (R)`	🔵 Blue `#3498DB`	**Ratio Locked (** `RpmPerRadS > 0` **):** Gear ratio locked after `>= 12` tight-cluster IQR samples; waiting for gear `N+1` ratio or top-gear WOT pull.
`N: (-)`	🔴 Red `#E74C3C`	**Unmapped / Paused:** Gear ratio not learned yet, memory reset, or game is paused / in garage menu (`IsRaceOn == 0`).

6. SIMHUB TELEMETRY PROPERTIES & ACTIONS REFERENCE (V1.1.0)

Property `[FH6SmartShift.*]`	Type	Description & Exact Behavior
`ActiveGameMode`	`string`	Returns `"FH6"` (Forza Horizon 6) or `"FM"` (Forza Motorsport) based on active SimHub game detection.
`CurrentCarKey` / `CurrentCarName`	`string`	Unique vehicle/tune ID (prefixed with `FM_` in Motorsport) / User-assigned car name from `CarList.json`.
`CurrentGearColor` / `CurrentGearStatusText`	`string`	Hex color (`#4CAF50` , `#3498DB` , `#F39C12` , `#E74C3C`) and status label of the currently engaged gear.
`ShiftTriggerRpm` / `MaxUsableRpm`	`double`	Shift Light trigger reduced by configurable driver reaction-time buffer (up to `1500 RPM` in G1–2, `600 RPM` in G3+).
`OptimalShiftRpm`	`double`	Physical unbuffered shift target RPM (`ShiftTriggerRpm` for `(P)` , `ShiftTriggerRpm - 50` for `(L)` , `actualLimiter - 250` for `(L?)`).
`DownshiftTriggerRpm`	`double`	Safe downshift RPM raised by braking (up to `+600 RPM`) and front dive (`+250 RPM`), capped by limiter ceiling.
`RevLimiterProximity`	`double`	`0.0-1.0` smooth LED progress toward buffered `ShiftTriggerRpm` with asymmetric EMA filtering.
`LedActiveCount` / `ShiftLightFlash`	`int/bool`	`0-12` active LED stage count with step hysteresis / `true` when all 12 LEDs lock at buffered shift point.
`IsShiftBeepEnabled`	`bool`	`true` when the F1 / GT3 audio shift beep (`2650 Hz` , `60 ms`) is enabled.
`Dragy_Last_0_100` / `Dragy_Best_0_100`	`double`	Last and Personal Best standing `0-100 km/h` acceleration time in seconds (sub-frame interpolated).
`Dragy_Last_100_200` / `Dragy_Best_100_200`	`double`	Last and Personal Best `100-200 km/h` time in seconds (standing or rolling pull).
`Dragy_Last_0_200` / `Dragy_Best_0_200`	`double`	Last and Personal Best standing `0-200 km/h` sprint time in seconds.
`Dragy_Status`	`string`	Live Dragy timer status string (e.g., `"READY (LAUNCH)"` , `"TIMING 0-100..."` , `"0-100: 3.12s"`).
`SlideSeverity` / `SlideTrend`	`double`	`0.0-1.0` integrated slip level above `15 km/h` / `-1.0..+1.0` slip time derivative (deepening vs. recovering).

Property `[FH6SmartShift.*]`	Type	Description & Exact Behavior
`SlideStateText` / `SlideBarColor`	`string`	`"▲ DEEPENING"`, `"● HOLDING"`, `"◄ RECOVERING"`, or `"GRIP"` / Hex color (`#E74C3C`, `#FF9900`, `#2ECC71`).
`LiveMappingProgress` / `EngineMapStatus`	`dbl/str`	Percentage (`0.0–100.0`) of populated active dyno bands / `"OK"` at `>= 92%`, else `"LEARN"`.
`RpmAtMaxPower` / `RpmAtMaxTorque`	`double`	RPM at which peak smoothed power and peak smoothed torque were recorded.
`IsDownshiftSafe` / `IsDownshiftSuggested`	`bool`	Safe downshift flag / Downshift suggested under braking or incline load (`_climbDownshiftActive`).
`IsGripLost` / `PowerOversteerWarning`	`bool`	Dashboard `SLIP` indicator (`> 1.25 G` or slip `> 0.30`) / Rear-axle WOT slip alert (`OVRST!`).
`BodySlipAngle` / `AeroBottomingWarning`	`dbl/bool`	Chassis slip angle in degrees / High-speed damper bottoming alert (`BOTTOM!` above `200 km/h`).
`Gear1Text..10Text` / `Gear1Color..10Color`	`string`	Status labels and Hex color codes for Gears 1–10.

⚡ SimHub Actions Reference

`FH6SmartShift.OpenDynoReport` (opens HTML Dyno, Dragy & Tuning Inspector) • `FH6SmartShift.ToggleShiftBeep` (toggles audio shift beep) • `FH6SmartShift.ResetDragyRecords` (clears current car Dragy PBs) • `FH6SmartShift.ResetGearRatios` (resets only ratios/shift points after tuning) • `FH6SmartShift.ResetLimitGears` (resets only `(L)` and `(L?)` gears) • `FH6SmartShift.ResetPowerMemory` (resets current car) • `FH6SmartShift.ResetAllCarsMemory` (full database wipe).

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